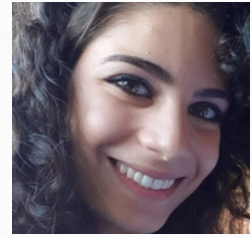


SOFIA FERAZZOLI



Ph.D. student in
Astronomy,
Astrophysics and
Space Science

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| [personal website](#) | [github](#) | [linkedin](#)

| [ORCID ID](#) | Date of birth 11 June 1997

ABOUT ME

Ph.D. student in Astronomy, Astrophysics and Space Science at Sapienza University of Rome, with a background in physics and observational cosmology. My work is mainly related to CMB polarization experiments, with experience in detector studies, data processing, simulations, and observing-strategy optimization within the QUBIC experiment. I am a part of QUBIC, ACT and MISTRAL collaborations.

I use Python for data analysis and scientific computing. I am interested in both experimental and computational aspects of astrophysics, and I have also taken part in outreach activities, developing communication skills alongside my research experience.

WORK EXPERIENCE

Sapienza University of Rome

City: Rome | **Country:** Italy

2025 - CURRENT **PhD Student - Astronomy, Astrophysics and Space Science (Cycle XLI)**

Supervised by Elia Stefano Battistelli

- Studying the polarization of the Cosmic Microwave Background through high-altitude ground-based experiments.
- Participating in international CMB polarization experiments, including QUBIC and Simons Observatory
- Contributing to instrument characterization, calibration, and performance monitoring during observational campaigns
- Developing and applying data analysis pipelines for map-making, power spectrum estimation, and systematic error characterization

EDUCATION

SAPIENZA UNIVERSITY OF ROME

City: Rome | **Country:** Italy

2025 - CURRENT **PhD degree in Astronomy, Astrophysics and Space Science (Cycle XLI)**

Field(s) of study: Astronomy,
Astrophysics and Cosmology

Level in EQF: EQF level 8

Thesis: Study of the polarization of the Cosmic Microwave Background from high altitude ground based experiments

Supervisor : Elia Stefano Battistelli

Co-supervisor : Silvia Masi

Studying CMB polarization through high-altitude ground-based experiments, contributing to instrument calibration and data analysis to probe cosmic inflation, lensing, reionization, and foreground contamination

2023 - 2025 **Master's degree in Astronomy and Astrophysics**

Field(s) of study: Astronomy,
Astrophysics and Cosmology

Level in EQF: EQF level 7

Final grade: 110/110 cum laude

Thesis: QUBIC: Detector Characterization through Map-Making Techniques and a Realistic Observation Plan

Supervisor : Elia Stefano Battistelli
Co-supervisor : Silvia Masi
External supervisor: Jean-Christophe Hamilton

My thesis focused on the QUBIC experiment, a ground-based experiment dedicated to observing the Cosmic Microwave Background's polarization. I developed an algorithm to identify the impact of cosmic rays on the detectors in order to characterize their time response. Using Monte Carlo simulations, I simulated the expected rate of cosmic rays at the QUBIC site. Finally, I developed an algorithm to determine the optimal observing schedule, maximizing the time on our primary scientific targets.

APRIL – MAY 2025 Master's thesis collaboration with APC Institute

Laboratoire Astroparticule & Cosmologie (APC), Université Paris Cité, Paris, France

Selection of the optimal observation windows to evaluate the scanning strategy for the QUBIC experiment.

2016 – 2022 Bachelor's Degree in Physics

Field(s) of study: Natural sciences, mathematics and statistics

Level in EQF: EQF level 6

Final grade: 96/110

Thesis: QUBIC: Cosmic Rays Detection

Supervisor : Elia Stefano Battistelli

Research activity carried out as part of my thesis within the QUBIC experiment, during which I joined the *QUBIC Collaboration*, focusing on the development of a Python algorithm to detect cosmic ray signatures in TES detector data and estimate detector time constants. The algorithm was later improved and expanded during my Master's studies.

Strong foundation in physics, mathematics, numerical methods, laboratory work, data analysis, C/Python programming, simulations, and scientific problem-solving.

San Bernardo Institute

City: Casamari (FR) | **Country:** Italy

2012 – 2016 Classic High School Diploma

Field(s) of study: Classical Studies, with a focus on Greek and Latin literature

Level in EQF: EQF level 4

Final grade: 100/100 cum laude

Focus on classical studies, particularly Greek and Latin literature, with complementary studies in languages, mathematics, and physics; served as student representative.

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s): English

LISTENING B2 READING B2 WRITING B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

- | | | |
|---------------|-------------|--------------------|
| • Linux/macOS | • Python, C | • Bash |
| • Jupyter lab | • Latex | • Microsoft Office |

TRAINING

SAPIENZA UNIVERSITY OF ROME

City: Rome | Country: Italy

2024 **Certification: Entry-Level Python Programming**

Python Institute

[PCEP-30-02] – Certification ID: RW46.6Csf.VLu6

Certified introductory training in Python, covering basic syntax, control flow, functions, data structures, and foundational coding practice.

2022 **Certificate of Attendance: Introduction to Python Programming**

CINECA

Certification ID: 446C53B0

Consolidated core Python skills, including syntax, control flow, functions, data structures, and file handling, with applications to scientific programming.

Basic Cybersecurity Training

INFN

Completed foundational training in information security, covering basic cybersecurity principles and safe digital practices.

2020–2021 **Basic Training Course in Sustainability Sciences**

Faculty of Civil and Industrial Engineering

Completed introductory training in sustainability sciences, covering fundamental concepts related to environmental, social, and technological sustainability.

OUTREACH

SAPIENZA UNIVERSITY OF ROME

City: Rome | Country: Italy

2026 **Introducing the TACOR telescope**

GUIDED ACTIVITY: Public science event “Notte Bianca dei Laboratori”

From screwdrivers to precision cosmology

POSTER: Student Orientation Poster Session event

Around the world in 8 CMB experiments

POSTER: She Rocks Science event

2025 **wikiHow to make a CMB experiment (with images)**

POSTER: She Rocks Science event

CONTRIBUTIONS UNIVERSITY OF BUENOS AIRES

City: Buenos Aires | Country: Argentina

2026 **Listening to the Early Universe: QUBIC and the Hunt for CMB B-Modes**

Journal club talk: An overview on the Cosmic Microwave Background Polarisation and on QUBIC characteristics as far as concern bolometric interferometry, cosmic rays detection, observation plan and detector characterization

During bilateral MAECI-Argentina program – 30 June 2026

MILANO-BICOCCA UNIVERSITY

City: Milan | Country: Italy

2024 **Q & U Bolometric Interferometer for Cosmology**

POSTER: An overview on QUBIC characteristics as far as concern bolometric interferometry and cosmic rays detection

CMB-CAL meeting, 4–8 November 2024

RESEARCH VISITS UNIVERSITY OF BUENOS AIRES

City: Buenos Aires | **Country:** Argentina

2026 Visiting Researcher – QUBIC Collaboration

MAECI (Ministero Affari Esteri Cooperazione Internazionale) funded Research Stay – QUBIC Collaboration.

One-month research stay involving on-site operations of the QUBIC instrument, observational activities, and data analysis in collaboration with the Argentinian team. Participated in instrumental operations at the QUBIC site.

During ilateral MAECI-Argentina program – June 2026

WORKSHOPS

2026 CENTER FOR COMPUTATIONAL ASTROPHYSICS, FLATIRON INSTITUTE

City: New York | **Country:** USA

- Flatiron Institute CMB Summer School. Intensive summer school on CMB data analysis and simulations, including hands-on exercises with contemporary CMB datasets and analysis tools.

SAPIENZA UNIVERSITY OF ROME

City: Rome | **Country:** Italy

- Workshop on Scientific Communication at INAF – Osservatorio Astronomico Romano (OAR)
 - Astrostatistics course for Astronomy
 - Machine Learning for Astronomy with hands on session
-

OTHER INTERESTS

Literature is my first passion. I deeply believe that words shape intentions and guide the way we experience life. I am especially drawn to English and Russian literature, whose landscapes and adventures have inspired me to *explore the world beyond the page*: places, histories, museums, and the traces of the past preserved in nature.

I enjoy feeling my body strong, focused, and under control, which has led me to develop a deep dedication to training, in particular swimming and *gym*. *Through challenges, I build my strength*.

The same attitude guides my relationship with coding: I find great excitement in going deeper into my code, debugging, improving, and trying to bring order to chaos.

The soundtrack of my life? The gentle sound of the flute, which I started playing as a child.

2009–PRESENT Playing the transverse flute as a hobby.

2007–2014 Attending swimming courses.

PUBLICATIONS

2026 Flux jump correction in the QUBIC TES data pipeline

Reference: B. Costanza et al., arXiv:2609.00249, 2026;

Authors: B. Costanza, L. M. Azzolina, C. G. Scoccola, N. M. Granese, E. Battistelli, J.-C. Hamilton, M. Piat, D. Prele, S. Torchinsky, I. Bekkaoui, A. Coppolecchia, P. De Bernardis, **S. Ferazzoli**, M. Gervasi, L. Gomez, S. Marwede, S. Masi, Y. Moulane, A. Paiella, M. Regnier, G. Romero, F. Sanchez, I. Queirolo

Journal/Archive: arXiv.

MISTRAL: optical alignment and beam optimization at the Sardinia Radio Telescope

Reference: E. Barbavara et al., SPIE Digital Library, Volume 14153,
DOI: <https://doi.org/10.1117/12.3103830>

Authors: E. Barbavara, A. Attoli, E. S. Battistelli, C. Bellucci, P. de Bernardis, F. Cacciotti, G. Carboni, E. Carretti, D. Cicalotti, F. Columbro, A. Coppolecchia, A. Cruciani, G. D'Alessandro, G. De Gasperis, M. De Petris, **S. Ferazzoli**, F. Govoni, G. Isopi, A. Ladu, L. Lamagna, P. Marongiu, D. Marsala, S. Masi, M. Murgia, A. Navarrini, A. Novelli, A. Occhiuzzi, A. Orlati, A. Paiella, G. Pettinari, F. Piacentini, M. Pili, G. Pisano, T. Pisanu, S. Poppi, L. Schirru, G. P. Vargiu, C. Volpe

Journal Name: SPIE Digital Library.

Design and construction of the QUBIC instrument mount

Reference: P. Strazzeri et al., SPIE Digital Library, Volume 14147,
DOI: <https://doi.org/10.1117/12.3103830>

Authors: P. Strazzeri, J. Alberro, E. S. Battistelli, A. Bottani, E. Carella, A. Coppolecchia, N. Del Castillo, G. de Innocenti, G. Domínguez, A. Etchegoyen, **S. Ferazzoli**, L. Ferreyro, J. Geria, D. Gorbeña, L. Grandsire, M. Hampel, C. Kristukat, O. Lang, F. Lazarte, A. Mancilla, S. Masi, J. Maya, J. Migliori, M. Mundo, H. Ocampo, A. Pantarotto, P. de Bernardis, A. Pereyra, A. Paiella, M. Platino, E. Rasztocky, A. Riello, P. Ringegni, C. Rodríguez, F. Rodríguez, G. Romero, D. Silva, A. Solís, L. Sucunza, C. Varela, G. Wasserman

Journal Name: SPIE Digital Library.

MISTRAL: first light from the Sardinia Radio Telescope and status of the technical commissioning

Reference: G. Isopi et al., SPIE Digital Library, Volume 14156,
DOI: <https://doi.org/10.1117/12.3103434>

Authors: G. Isopi, A. Attoli, E. Barbavara, E. S. Battistelli, C. Bellucci, P. de Bernardis, F. Cacciotti, G. Carboni, E. Carretti, D. Cicalotti, F. Columbro, A. Coppolecchia, A. Cruciani, G. D'Alessandro, G. De Gasperis, M. De Petris, **S. Ferazzoli**, F. Govoni, A. Ladu, L. Lamagna, P. Marongiu, D. Marsala, S. Masi, M. Murgia, A. Navarrini, A. Novelli, A. Occhiuzzi, A. Orlati, A. Paiella, G. Pettinari, F. Piacentini, M. Pili, G. Pisano, T. Pisanu, S. Poppi, L. Schirru, G. P. Vargiu, C. Volpe

Journal Name: SPIE Digital Library.

Performance of the QUBIC CMB telescope's detectors

Reference: S. Marwede et al., SPIE Digital Library, Volume 14156,
DOI: <https://doi.org/10.1117/12.3102976>

Authors: S. Marwede, C. O'Sullivan, A. B. Flood, D. Gayer, M. Piat, S. A. Torchinsky, E. Barbavara, E. S. Battistelli, P. de Bernardis, M. Bianchessi, F. Columbro, A. Coppolecchia, B. Costanza, **S. Ferazzoli**, L. Ferreyro, A. Fuster, B. García, M. Gervasi, L. Gomez Bachar, L. Grandsire, M. Hampel, G. Isopi, A. Mancilla, S. Masi, A. Mennella, N. Mirón-Granese, H. Ocampo, A. Paiella, D. Prêle, I. Queirolo, E. Rasztocky, C. A. Rodríguez, F. Rodríguez, G. E. Romero, F. Sánchez, C. Scóccola, A. Solís, A. D. Supanitsky, A. Vazquez, M. Zannoni

Journal Name: SPIE Digital Library.

The effect of QUBIC beam uncertainty on the reconstruction of the CMB B-mode power spectrum

Reference: A. B. Flood et al., SPIE Digital Library, Volume 14156,
DOI: <https://doi.org/10.1117/12.3102976>

Authors: A. B. Flood, C. O'Sullivan, M. L. Gradziel, D. Gayer, S. Marwede, C. G. Scóccola, E. S. Battistelli, P. de Bernardis, F. Columbro, A. Coppolecchia, B. Costanza, **S. Ferazzoli**, B. Garcia, M. Gervasi, A. Huchet, T. Laclavere, S. Masi, A. Mennella, N. Mirón-Granese, A. Paiella, M. Piat, E. Rasztocky, M. Regnier, G. E. Romero, S. A. Torchinsky, M. Zannoni

Journal Name: SPIE Digital Library.

QUBIC: an algorithm for detecting cosmic rays

Reference: S. Ferazzoli et al., Frontiers in Astronomy and Space Sciences, 12:1659952, 2026

Data Analysis (QUBIC): Developed a Python algorithm to detect above-noise cosmic ray signatures in the time-ordered data of TES bolometers for the QUBIC experiment.

Data Processing: Applied the algorithm to datasets acquired during the 2022 QUBIC testing campaign in Salta, Argentina, including pre-processing, threshold-based event selection, and statistical validation of candidates.

Authors: S. Ferazzoli, E. S. Battistelli, S. Masi, E. Barbavara, A. Coppolecchia, B. Costanza, P. De Bernardis, G. De Gasperis, M. Gervasi, J.-Ch. Hamilton, N. Miron Granese, G. Isopi, C. O'Sullivan, A. Paiella, M. Piat, F. Sánchez, C. Scóccola, S. A. Torchinsky and M. Zannoni

Journal Name: Frontiers in Astronomy and Space Sciences.

Spectral imaging with QUBIC: building astrophysical components from Time-Ordered-Data using Bolometric Interferometry

Reference: M. Regnier et al., JCAP04(2026)018.

DOI: 10.1088/1475-7516/2026/04/018

Authors: M. Regnier, T. Laclavère, J.-Ch. Hamilton, E. Bunn, V. Chabirand, P. Chanial, L. Goetz, L. Kardum, A. Huchet, P. Masson, N. Miron Granese, C. G. Scóccola, S. A. Torchinsky, E. Battistelli, M. Bersanelli, F. Columbro, A. Coppolecchia, B. Costanza, P. De Bernardis, G. De Gasperis, S.

Ferazzoli, A. Flood, K. Ganga, M. Gervasi, L. Grandsire, E. Manzan, S. Masi, A. Mennella, L. Mousset, C. O'Sullivan, A. Paiella, F. Piacentini, M. Piat, L. Piccirillo, E. Rasztocky, M. Stolpovskiy and M. Zannoni

Journal Name: Journal of Cosmology and Astroparticle Physics.

Spectral Imaging with QUBIC: building frequency maps from Time-Ordered-Data using Bolometric Interferometry

Reference: M. Regnier et al., JCAP05(2026)032.

DOI: 10.1088/1475-7516/2026/05/032

Authors: M. Regnier, T. Laclavère, J.-Ch. Hamilton, E. Bunn, V. Chabirand, P. Chanial, L. Goetz, L. Kardum, A. Huchet, P. Masson, N. Miron Granese, C.G. Scóccola, S.A. Torchinsky, E. Battistelli, M. Bersanelli, F. Columbro, A. Coppolecchia, B. Costanza, P. De Bernardis, G. De Gasperis, S.

Ferazzoli, A. Flood, K. Ganga, M. Gervasi, L. Grandsire, E. Manzan, S. Masi, A. Mennella, L. Mousset, C. O'Sullivan, A. Paiella, F. Piacentini, M. Piat, L. Piccirillo, E. Rasztocky, M. Stolpovskiy and M. Zannoni

Journal Name: Journal of Cosmology and Astroparticle Physics.

2025 The use of physical optics and machine learning in modelling the QUBIC beam pattern

Reference: A. B. Flood et al., Proceedings of SPIE, Volume 13623 (2025).

DOI: 10.1117/12.3064328

Authors: A. B. Flood, C. O'Sullivan, M. L. Gradziel, D. Gayer, S. Marwede, J.-Ch. Hamilton, S. Torchinsky, T. Laclavère, L. Kardum, M. Regnier, E. S. Battistelli, P. de Bernardis, B. Costanza, S. Ferazzoli, M. Gervasi, S. Masi, N. Miron-Granese, L. Mousset, C. Scóccola and M. Zannoni

Journal/Proceedings: Proceedings of SPIE

I authorize the processing of my personal data contained in the CV pursuant to art. 13 of Legislative Decree no. 196 of June 30, 2003 – "Personal Data Protection Code" and art. 13 of GDPR 679/16 – "European Regulation on Personal Data Protection."

Rome, May 2026

Sofia Ferazzoli

Sofia Ferazzoli